# *Hire Your Anthropologist!*
# Towards Cultural Benchmarks Through an Anthropological Lens

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## Abstract

This position paper critically examines existing benchmarks for evaluating culture in large language models (LLMs) through the framework of cultural anthropology, guided by Russell's Handbook for Methods in Cultural Anthropology (Bernard and Gravlee, 2015). We collaborate with an anthropologist on the paper and advocate for the collaboration with anthropologists because anthropologists have been specifically concerned with the culture question and its shifts and development since the early 20th century. Anthropology is also the most interpretative of the social sciences and thus working with anthropological methods works well side by side to quantitative methods. We analyze 17 widely used benchmarks, highlighting their strengths and limitations in capturing culture's dynamic, context-dependent nature. Our analysis reveals that these benchmarks often reduce culture to a static, easily quantifiable construct, however overlooking the philosophical and interpretative nuances that underpin cultural diversity. To address these gaps, we propose a set of recommendations for redesigning benchmarks, including the integration of ethnographic methods, intersectional frameworks, and cross-cultural datasets. By re-imagining cultural benchmarks, we seek to enable a more nuanced understanding of how culture is constructed and negotiated within communities, ultimately guiding the development of LLMs that are more culturally aware, inclusive, and adaptable. Accordingly, we provide recommendations and we propose a way forward for improving cultural benchmarks through interdisciplinary collaboration between AI researchers and anthropologists, fostering methodologies that bridge computational techniques with deep cultural insights.

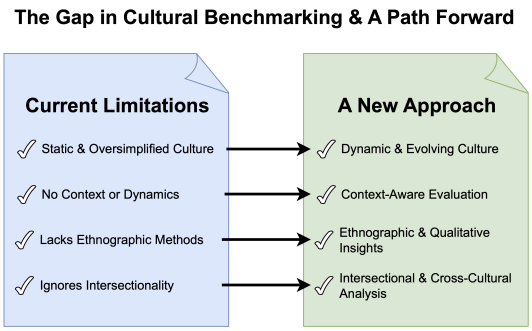


Figure 1: Rethinking cultural benchmarks through an anthropological lens.

## 1 Introduction

In this paper, we review benchmarks designed to evaluate cultural dimensions in large language models (LLMs), drawing on methods from cultural anthropology. Evaluating these cultural dimensions is increasingly crucial as LLMs begin to shape societal interactions, inform decision-making, and influence the dissemination of knowledge. We base our review on methods drawn from cultural anthropology. For the purposes of this paper, we rely on Russell's "Handbook for Methods in Cultural Anthropology," (Bernard and Gravlee, 2015). Anthropology is particularly well-suited for this endeavor because it is the discipline tasked with understanding the concept of culture for the past six decades. Anthropologists have asked questions examining the concept of cultures; They have studied different cultures around the world, conducted comparative analysis, as they shaped the field of cultural studies.

Importantly, methods in anthropology are both practical and philosophical. The way researchers collect data influences the data itself. By immersing themselves in local communities, anthropologists not only document facts but also introduce their own subjectivities into the resulting ethnographies. This realization inspired the "writing cul-

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ture" turn in the 1980s, a seminal book arguing that ethnographic facts do not exist independently of the researcher's lens (Clifford and Marcus, 1986). This philosophical shift led to an interpretative turn in anthropology, emphasizing that social concepts are continuously produced and negotiated rather than existing as static entities.

For quantitative researchers, interpretative methods present challenges because they are difficult to quantify and operationalize. However, with the advent of LLMs and the breadth of data they process, we can explore ways to incorporate interpretative approaches into LLMs' understanding of cultural diversity. This requires benchmarks that go beyond factual accuracy to capture the fluidity and contextual nature of culture.

## 2 Methods

In this study, we use the term "cultural benchmarks" to describe evaluation datasets and metrics designed to measure elements of cultural understanding in LLMs. These benchmarks aim to capture how LLMs interpret, generate, and align with cultural norms, values, and biases across diverse contexts. Our selection criteria focused on benchmarks that are designed to explicitly or implicitly assess cultural awareness, social norms, and ethical reasoning. We find that these benchmarks – despite being important initial steps in evaluating LLMs' understanding of culture and bias – tend to treat culture as a static, operationalizable concept that can be enumerated through checklists of facts. The challenge for these benchmarks lies in capturing the nuanced and context-dependent nature of social behavior. Culture, bias, and values are increasingly used as buzzwords for better model alignment. However, we wonder to whom are LLMs aligning? And, how can we train models to philosophically engage with the complexities of social experience?

### 2.1 Benchmarks

To analyze these benchmarks through an anthropological lens, we employ methods such as comparative cultural analysis, contextual interpretation, and intersectional evaluation. Drawing from ethnographic practices, we examine the benchmarks within their sociocultural contexts, identifying how they reflect or obscure cultural diversity and variability. We also apply cultural relativism to avoid imposing universalistic interpretations on diverse cultural phenomena, ensuring that each benchmark is evaluated with sensitivity to its unique framework and purpose.

**Criteria of Selection** The selection of cultural benchmarks for this study was guided by their relevance in evaluating the engagement of LLMs with cultural norms, biases, and social contexts. We prioritized benchmarks that met the following criteria:

1. **Explicit or Implicit Cultural Assessment:** Benchmarks that evaluate LLMs' understanding of cultural norms, biases, and ethical reasoning explicitly or implicitly;
2. **Breadth of Contexts:** Benchmarks designed to test LLMs across diverse scenarios, encompassing both universal and culturally specific contexts.
3. **Current Relevance:** These benchmarks are frequently cited in AI research, addressing issues such as social biases, ethical reasoning, and multilingual understanding.
4. **Applicability to Anthropology:** Benchmarks whose methodologies align with anthropological principles, particularly those emphasizing cultural variability and context-dependent norms.

We reviewed 17 cultural benchmarks based on their relevance to evaluating LLMs' engagement with cultural norms, biases, and social contexts. A brief overview of each benchmark, including its design focus and limitations, is provided in Appendix A.1. These benchmarks include Social Chemistry 101 (Forbes et al., 2020), BBQ (Parrish et al., 2022), COCO-LM (Meng et al., 2021), CrOWs Pair (Nangia et al., 2020a), and others.

Across these benchmarks, we observe a general trend of treating culture as static and operationalizable, rather than dynamic and context-dependent. In the following sections, we analyze their methodologies and shortcomings through an anthropological lens.

## 3 Analysis

To evaluate the effectiveness of current cultural benchmarks, we analyzed their methodologies and outcomes through an anthropological lens, emphasizing contextual representation, bias detection, cultural models, factual consistency, and social norms. The following analysis highlights strengths, limi-

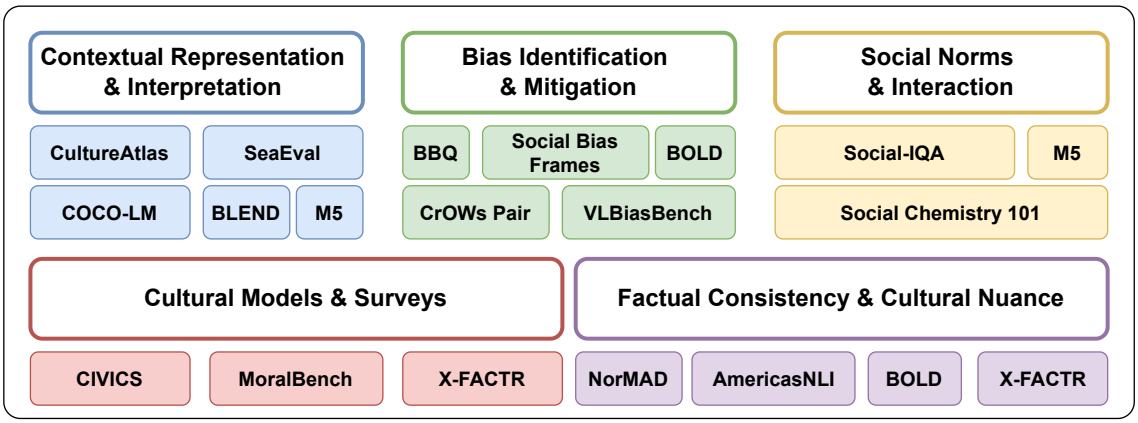


Figure 2: The cultural benchmarks analyzed in this work.

tations, and opportunities for improvement, while illustrating specific examples from each dataset.

### 3.1 Contextual Representation and Interpretation

Benchmarks such as CultureAtlas (Fung et al., 2024), SeaEval (Wang et al., 2024a), COCO-LM (Meng et al., 2021), BLEND (Myung et al., 2024), and M5 (Schneider and Sitaram, 2024) attempt to evaluate the ability of LLMs to navigate varied cultural contexts.

**CultureAtlas** assesses the adaptability of LLMs in recognizing different conversational norms across cultures. While it provides valuable insight into cross-cultural communication, it often reflects dominant cultural frameworks, which may overshadow underrepresented dialects and communication patterns. To assess such effectin CULTUREATLAS, we conduct a quantitative analysis of sub-country level cultural coverage across 102 countries, as reported in Table 9 of the original paper. We define two key metrics: *Cultural Representation Rate* (CRR) and *Cultural Disparity Ratio* (CDR), and complement these with an analysis of zero-coverage countries.

$$\text{CRR}_i = \frac{\text{Sentences}_i}{\sum_j \text{Sentences}_j}$$

$$\text{CDR} = \frac{\sum_{i \in \text{Top-5}} \text{CRR}_i}{\sum_{j \in \text{Bottom-5 (non-zero)}} \text{CRR}_j}$$

We compute a **Cultural Disparity Ratio (CDR)** of 169.2, indicating that the top five most-represented countries (United States, India, Italy, China, and Japan) are collectively over 169 times more represented than the bottom five nonzero countries (Liechtenstein, Liberia, Tuvalu, Belize, and Andorra) combined. In addition, we calculate the **Zero Coverage Rate**, defined as the proportion of countries with no sub-country level cultural knowledge assertions. Out of 102 countries, **28** have zero entries, corresponding to a **27.5%** zero-coverage rate. Notably, this group includes numerous Global South and small island nations (e.g., Haiti, Guatemala, Tonga), highlighting significant geographic and ethnolinguistic underrepresentation in the dataset. This also aligns with our CDR results in appendix 1.

**SeaEval** Although SeaEval commendably seeks to assess multilingual models' cultural comprehension abilities by manually constructing four datasets (US-Eval, SG-Eval, CN-Eval, and PH-Eval) across distinct regions, the selection of these cultural contexts appears somewhat arbitrary and lacks ethnographic grounding. The rationale for choosing these specific region is not clearly articulated. As a result, the benchmark risks oversimplifying the vast cultural heterogeneity within and across these areas. For example, Singapore and the Philippines each encompass diverse linguistic, ethnic, and socio-economic groups, whose cultural values cannot be meaningfully captured through a single monolingual dataset, especially when some of the languages are not native. A more ethnographically informed approach—grounded in fieldwork, sociolinguistic variation, and subcultural differentiation—would enable the construction of datasets that better reflect the fluid, situated, and contested nature of cultural knowledge, thereby offering a more rigorous and realistic evaluation of models' cross-cultural abilities.

**Successful Example** BLEND focuses on synthesizing contextually rich narratives from diverse sources, offering a more holistic approach to cultural representation. M5 evaluates multimodal contexts in diverse languages and cultures, providing an opportunity to analyze how LLMs interact with cultural norms across text and images. These benchmarks highlight the importance of dynamic contextual assessments but remain constrained by rigid cultural assumptions.

### 3.2 Bias Identification and Mitigation

BBQ (Parrish et al., 2022), CrOWs Pair (Nangia et al., 2020a), Social Bias Frames (Sap et al., 2020), BOLD (Dhamala et al., 2021), and VLBiasBench (Wang et al., 2024b) focus on identifying biases within LLMs, particularly in relation to protected identities.

**BBQ** provides a robust framework for evaluating how question-answering models reinforce explicit social stereotypes.However, its templated and binary format often fails to capture the dynamic, context-sensitive nature of sociocultural biases, which evolve over time and across settings. These formats typically rely on static prompt structures, which may insufficiently reflect the nuanced and intersectional ways bias manifests in real-world interactions. By treating bias as a binary attribute that is or is not associated with a particular identity category, benchmarks risk essentializing identity itself, ignoring how someone can be simultaneously privileged and marginalized depending on their cultural, national, or situational context(Crenshaw, 2013).

**CrOWs Pair** assesses whether LLMs assign higher probabilities to stereotype-based sentences over neutral alternatives. While CrowS-Pairs improves over templated benchmarks through its use of crowd-sourced stereotype pairs, it still operates under a fixed and binary framing of bias: each example is labeled with a single stereotype (e.g., race, religion), and assumes that biases are universally interpretable and stable across contexts.This design deliberately avoids representing intersectional or overlapping biases, which are essential to understanding how identity and discrimination manifest in real life (Crenshaw, 2013). Examples with annotator disagreement about the type of bias are discarded, treating disagreement as noise rather than as evidence of context-sensitive or culturally variable interpretations of bias(Pavlick and Kwiatkowski, 2019).

**Social Bias Frames** provides a nuanced analysis of implicit bias by evaluating the social implications of different textual interpretations.While Social Bias Frames advances the field by modeling pragmatic bias implications through structured and free-text annotations, its framing remains largely unidimensional. That is, it treats demographic group categories (e.g. gender, race, religion) as independent variables, failing to account for the interactive effects of multiple marginalized identities—a core tenet of intersectionality. This omission not only limits the system's cultural fidelity but also risks erasing compound discrimination, where harm emerges not from any single identity but from their interdependence(Sue, 2010).

**Positive example** BOLD stands out in evaluating bias in LLMs because it focuses on open-ended text generation, capturing how stereotypes emerge organically through subtle phrasing and narrative framing—unlike the cloze-style (BBQ) or binary sentence pair (CrowS) formats that constrain model outputs to predefined choices. BOLD spans a wider range of domains, including underexplored areas like profession and political ideology, and emphasizes implicit bias over explicit preferences, offering greater ecological validity and alignment with real-world usage scenarios. Its naturalistic prompts and scale (over 23k examples) make it a more realistic and nuanced benchmark for assessing systemic stereotype reinforcement in generative AI.

VLBiasBench exemplifies a more comprehensive approach to bias identification by extending beyond language-only evaluations and incorporating multimodal contexts, where both images and text shape interpretation. By analyzing responses to a large-scale set of synthetic visual prompts across a diverse set of social and intersectional categories, it reveals how biases can emerge not only from wording but also from visual representation and their interplay. This multimodal scope allows for a richer, more realistic understanding of how AI systems reflect and reinforce societal stereotypes in everyday applications, capturing forms of bias that purely linguistic datasets may overlook.

### 3.3 Cultural Models and Surveys

Benchmarks such as CIVICS (Pistilli et al., 2024), MoralBench (Ji et al., 2024). While these benchmarks provide useful insights, they often rely on static moral principles that are potentially undergeneralized or overgeneralized rather than accom-

modating cultural pluralism.

**CIVICS** evaluates how models respond to sociopolitical issues like LGBTQ+ rights and immigration through a structured approach, but despite including five languages (English, French, German, Italian, and Turkish), its cultural coverage remains severely limited to primarily Western and European perspectives. This narrow sample excludes vast regions of Asia, Africa, Latin America, and Indigenous communities worldwide, risking evaluation bias as models' responses to ethical topics are judged mainly against Western liberal democratic norms rather than diverse global value systems, ultimately imposing universal moral frameworks that don't account for cultural relativism and the varied ethical reasoning that shapes these issues across different societies.

**MoralBench** tests LLMs on moral dilemmas, assumes that ethical reasoning is a fixed construct rather than one shaped by regional traditions and values. This approach faces significant limitations when attempting to ground morality using Moral Foundations Theory (MFT) in multicultural contexts. The fundamental limitation lies in MFT's Western-centric development. While MFT identifies their dimensions as universal moral foundations, these categories emerged primarily from research with WEIRD (Western, Educated, Industrialized, Rich, Democratic) populations.

### 3.4 Factual Consistency and Cultural Nuance

NorMAD (Rao et al., 2024), Americas NLI (Ebrahimi et al., 2022), BOLD (Dhamala et al., 2021), and X-FACTR (Jiang et al., 2020) assess the ability of LLMs to provide factually accurate information while considering cultural interpretations.

**NorMAD** tests factual information across different cultural perspectives, ensuring that models do not default to dominant historical narratives.However, its dependence on sources such as the Australian Cultural Atlas risks reproducing Western-centric epistemologies. By presenting culture as static and monolithic, it echoes longstanding critiques in postcolonial and decolonial scholarship that warn against essentializing non-Western societies through external lenses (Spivak, 1988; Smith, 1999). The absence of mechanisms to validate cultural norms through indigenous voices or contested local narratives may lead to the uncritical reinforcement of colonial knowledge hierarchies (Mudimbe, 1988), allowing LLMs to default to dominant global narratives. As Bhambra (Bhambra, 2014) argues, such epistemic asymmetries persist when knowledge systems from the Global South are represented without reflexivity or reciprocity, raising critical concerns about the benchmark's capacity to foster true cultural pluralism.

**AmericasNLI** evaluates multilingual LLM, particularly in indigenous languages, expanding linguistic representation. However, it falls significantly short of incorporating indigenous epistemologies. This limitation stems largely from its methodology: the data set is primarily translated from XNLI, introducing many concepts alien to Indigenous worldviews. The direct translation approach imports foreign conceptual frameworks, metaphors, and reasoning patterns that often clash with Indigenous knowledge systems. These imposed Western linguistic structures and cultural assumptions fail to capture the nuanced relationships with land, community, and spirituality central to many Indigenous epistemologies. Consequently, while the dataset increases language coverage, it evaluates LLM performance through a lens that fundamentally misaligns with Indigenous ways of knowing, limiting its effectiveness in capturing the depth and complexity of cultural narratives rooted in these distinct epistemological traditions.

**X-FACTR** 's strength lies in its multilingual knowledge validation, which offers a broad evaluation in 23 typologically diverse languages, but it could be further improved by incorporating culturally specific interpretations of facts, such as recognizing that factual importance, lexical choice, and associative prominence differ across languages due to sociocultural grounding, thus shifting the evaluation from mere linguistic variation to deeper cultural cognition and semantic structure, which is critical for fair and representative benchmarking in low-resource or culturally distinct languages.

**Good examples** BOLD evaluates how biases appear in open-ended text generation, highlighting where LLMs reinforce stereotypes through subtle wording choices.Unlike static translation-based benchmarks or culture-as-fact evaluations that risk reifying Western-centric epistemologies, this approach captures how language models reinforce social biases through subtle generative associa-

tions—revealing not just what models know, but how they express it. By sampling prompts from naturally occurring, crowd-authored content and evaluating outputs with a suite of metrics (e.g., sentiment, toxicity, gender polarity, psycholinguistic norms), it surfaces nuanced biases across identity categories without relying on predefined labels or forced-choice tasks. Its open-ended structure allows for emergent patterns—such as disproportionate negativity toward marginalized groups or occupational gender skew—providing a richer diagnostic lens into model behavior and its sociolinguistic consequences.

### 3.5 Social Norms and Interaction

Benchmarks such as Social IQA (Sap et al., 2019), Social Chemistry 101 (Forbes et al., 2020), and M5 (Schneider and Sitaram, 2024) measure the ability of LLMs to navigate implicit social norms.

**Social IQa** is a prominent benchmark designed to evaluate models' capacity for social and emotional commonsense reasoning, particularly in navigating everyday interpersonal scenarios. However, as the dataset draws heavily from crowdsourced annotations sourced primarily within Western sociocultural contexts, it inherently encodes assumptions aligned with individualist norms (e.g., personal motivation, self-driven action) while underrepresenting collectivist and interdependent frameworks prevalent in many non-Western cultures. This echoes broader concerns raised in cross-cultural psychology, where cognitive reasoning styles and social expectations are known to vary systematically across societies (Markus and Kitayama, 1991). Consequently, while SOCIAL IQA offers valuable insight into models' understanding of Western social reasoning, its utility as a globally representative benchmark remains limited.

**Social-Chemistry101** Although Social-Chemistry-101 provides a rich 292k-item inventory of everyday "rules-of-thumb," its authors explicitly limit both prompts and annotators to "English-speaking cultures represented within North America," drawing every situation from U.S. sources (r/AITA, r/confessions, ROCStories) and noting that 94% of workers have lived in the United States for 10+ years (Forbes et al., 2020). A Chinese–English replication therefore treats the original corpus as a baseline of American folk ethics and builds new data precisely to capture norms, e.g., filial piety, face maintenance, that the US centric set omits.(CH-Wang et al., 2023)

**M5** , which evaluates LLM on diverse vision-language tasks, provides an opportunity to test cultural norms in multimodal contexts, but it still struggles to account for cultural changes over time. e. That is, its benchmark reflects a snapshot of cultural representation but lacks mechanisms for evaluating how LMMs respond to evolving cultural norms or temporal shifts in visual semiotics, language usage, or sociocultural values. Furthermore, the benchmark exposes significant disparities between high- and low-resource languages, with many models performing near or below random baseline on new tasks like Visio-Linguistic Outlier Detection (VLOD) in non-Western languages. This highlights the urgent need not only for broader cultural inclusivity but also for temporally adaptive evaluation frameworks that can assess whether models evolve with societal shifts—an area that remains underexplored in M5's current design.

These benchmarks provide valuable information about social interactions but must expand their datasets to capture a broader range of cultural behaviors.

### 3.6 Successful Examples of Contextual Use

Some benchmarks demonstrate strong methodologies in representing cultural variability and adaptability. SeaEval (Wang et al., 2024a) and Culture-Atlas (Fung et al., 2024) effectively assess multilingual cultural reasoning and conversational norms, making them useful for testing cultural adaptability in cross-linguistic settings. NorMAD (Rao et al., 2024) provides a valuable framework for assessing factual consistency in culturally specific contexts, ensuring that historical and factual knowledge is not skewed toward dominant narratives. CIVICS (Pistilli et al., 2024) successfully integrates socially relevant and value-laden prompts to measure how LLMs handle contentious cultural issues across multiple societies. Finally, Social Chemistry 101 (Forbes et al., 2020), despite its western-centric data source, offers a structured categorization of social norms, which could be enhanced by expanding its dataset to include culturally diverse social customs.

## 4 Recommendation

To improve the ability of benchmarks to measure cultural understanding in LLMs, we propose the

following recommendations, connecting themes across the reviewed benchmarks:

**Adopt Dynamic, Contextual Frameworks** Benchmarks should move beyond static lists of norms and biases by adopting dynamic frameworks that reflect the fluidity of culture. Social Chemistry 101 and BBQ could benefit from incorporating real-world, context-sensitive data to better capture how norms and biases shift across different scenarios. For example, rather than focusing solely on cataloging rules, benchmarks could introduce adaptive tests that change based on regional and temporal variations in social norms.

**Incorporate Ethnographic Methods** Ethnographic fieldwork emphasizes the situated nature of cultural knowledge Geertz, 1973. Benchmarks like COCO-LM (Meng et al., 2021) and CrOWs Pair (Nangia et al., 2020b) should integrate findings from long-term ethnographic research to provide richer, more nuanced contexts. This would enable LLMs to process and generate culturally relevant information that mirrors lived human experiences.

**Expand Cross-Cultural and Multilingual Datasets** Since culture is inherently diverse, benchmarks should include a wider array of cultural and linguistic data. Additionally, these benchmarks could benefit from ongoing updates to reflect contemporary cultural shifts for example with migratory flows that happen between countries with ongoing wars, new geopolitical shifts and similar examples. Enhance Bias Detection through Intersectional Analysis Bias benchmarks such as Social Bias Frames (Sap et al., 2020) and BBQ (Parrish et al., 2022) should incorporate intersectional frameworks Crenshaw, 1989 that account for the overlapping nature of biases (e.g., race, gender, class). This approach recognizes that biases cannot be fully understood in isolation but must be analyzed within broader socio-cultural dynamics.

**Foster Interpretative Flexibility** Benchmarks should allow for multiple valid interpretations of social norms and biases. Social IQA (Sap et al., 2019) and BLEND (Myung et al., 2024), for example, could introduce scoring mechanisms that reward models for offering diverse contextually appropriate responses. This would encourage LLMs to engage with cultural complexities rather than offering reductive answers.

**Collaborate with Social Scientists and Anthropologists** Developing benchmarks that truly capture cultural understanding requires interdisciplinary collaboration. By working closely with anthropologists, sociologists, and other social scientists, AI researchers can design benchmarks that are methodologically robust and culturally sensitive. Evaluate Long-Term Model Adaptability Benchmarks should assess not only how well models perform on static tests, but also how they adapt over time. This could involve longitudinal studies where models are periodically tested with updated datasets to track their learning and adaptability to cultural changes.

## 5 Conclusion

This paper critically examined cultural benchmarks in LLMs through an anthropological lens, finding that while they provide insights into bias detection and ethical reasoning, they often treat culture as static and quantifiable rather than fluid and context-dependent. Our analysis of 17 benchmarks revealed this reductionist approach fails to capture how individuals and communities collaboratively construct meaning. Rather than defining culture as fixed, we should focus on how people conceptualize it as a dynamic process shaped by communal meanings. Our recommendations include integrating ethnographic methods, expanding cross-cultural datasets, improving bias detection with intersectional analysis, and fostering interdisciplinary collaboration between AI researchers and social scientists.

## Limitations

While this study provides a comprehensive critique of existing cultural benchmarks and offers concrete recommendations, it is not without limitations. One significant constraint lies in the scope of benchmarks analyzed. Although we reviewed 17 cultural benchmarks, our selection was necessarily limited to publicly available datasets and widely recognized frameworks. As a result, other relevant benchmarks, particularly those that are emerging or proprietary, may provide additional insights that were not covered in this work. Notably, we did not review all benchmarks working to tackle these issues. Several key datasets, such as the PRISM alignment dataset (Kirk et al., 2025), Value Kaleidoscope (Sorensen et al., 2024), and Social Norms in Cinema (**?**), have made significant strides toward overcoming anglocentric biases and embrac-

ing pluralistic cultural perspectives. However, we excluded them from our analyses to maintain a bird's-eye view focused on categorizing broader types of critiques across benchmarks, rather than conducting an exhaustive dataset inventory.

Another limitation is the dependence on existing data sources, many of which are derived from online platforms such as Reddit, Wikipedia, and social media. While these datasets are valuable for large-scale benchmarking, they often reflect dominant cultural narratives while underrepresenting marginalized voices. This bias in data collection influences how culture is defined and measured within LLMs, potentially reinforcing stereotypes and excluding diverse cultural perspectives. Our analysis acknowledges these biases but does not directly address alternative data collection methods that could better capture underrepresented viewpoints.

Additionally, operationalizing interpretative methods within computational benchmarking presents significant challenges. While this paper advocates for integrating ethnographic and interpretative approaches, implementing these methodologies in AI evaluation remains an open research problem. Unlike traditional quantitative metrics, interpretative cultural assessments rely on subjective, evolving, and context-dependent meanings, making them difficult to standardize. The lack of clear, automated metrics for evaluating nuanced cultural interpretations poses a major hurdle in applying these methods effectively at scale. Interdisciplinary collaboration is another critical challenge. While we emphasize the need for AI researchers and anthropologists to work together in designing more culturally informed benchmarks, institutional and methodological barriers often hinder such efforts. Differences in research priorities, epistemological approaches, and disciplinary methodologies create obstacles to productive dialogue and integration. Without structured frameworks to facilitate interdisciplinary collaboration, efforts to improve cultural benchmarking risk remaining fragmented or underutilized. Furthermore, the complexity of evaluating cultural understanding in LLMs presents additional difficulties. Our proposed pipeline introduces a novel approach by incorporating ethnographic insights and community-based assessments, but its feasibility on a large scale remains uncertain. Community-driven evaluations, while essential for capturing cultural authenticity, require extensive resources, ethical oversight, and iterative validation. Implementing such methods across a diverse range of cultural contexts is a resource-intensive endeavor that may limit their practical adoption in AI research and industry settings.

Finally, our study primarily focuses on benchmarking methodologies rather than assessing the real-world impact of cultural benchmarks on LLM applications. While improving benchmarks is a crucial step toward cultural sensitivity in AI, future research should investigate how these advancements translate to tangible improvements in AI-human interactions across various domains, including education, governance, healthcare, and cross-cultural communication. Without real-world deployment analysis, the long-term effectiveness of improved cultural benchmarks remains an open question.

Despite these limitations, this work serves as an important step toward rethinking cultural benchmarking in AI. By advocating for more adaptable, interdisciplinary, and community-driven approaches, we hope to inspire future research that refines evaluation methodologies, expands cultural datasets, and explores new ways to operationalize cultural fluidity in computational settings.

## Ethical Statement

This position paper advocates for a rethinking of cultural benchmarks in Large Language Models (LLMs) through an anthropological lens, a perspective that introduces several ethical considerations regarding the framing, methodology, and potential impact of our analysis.

The issue of intellectual ownership and participatory ethics is also relevant to this discussion. By advocating for the incorporation of ethnographic data and lived experiences into AI benchmarks, this paper indirectly raises concerns about how cultural knowledge is sourced and used. Ethical AI research must avoid extractive approaches that commodify cultural knowledge without consent or compensation. The recommendations presented in this paper should be interpreted as a call for ethically co-designed benchmarks rather than an endorsement of appropriating cultural insights without the involvement of the communities they pertain to.

Furthermore, this paper discusses bias and misrepresentation in AI but operates within the constraints of currently available benchmarks. The selection of 17 cultural benchmarks, while intended to be

comprehensive, is still shaped by the availability of research, the predominance of English-language sources, and our own methodological choices. Any analysis of AI benchmarks carries the risk of reinforcing certain academic and institutional perspectives while overlooking others. Future interdisciplinary research should seek to include voices from non-Western epistemologies, Indigenous knowledge systems, and community-driven AI governance models.

Overall, this paper seeks to approach cultural benchmarking in AI with ethical responsibility, advocating for methodologies that are both rigorous and respectful of the diverse human experiences they seek to represent. However, the very act of critiquing, selecting, and proposing cultural benchmarks must itself remain subject to ethical scrutiny, ensuring that the pursuit of cultural sensitivity in AI does not inadvertently replicate the biases and epistemic imbalances it aims to address

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# A Appendix

## A.1 Benchmark Summaries

Below we provide short descriptions of the 17 cultural benchmarks reviewed in this study:

**Social Chemistry 101** Forbes et al. (2020) A dataset derived from Reddit and other online sources compiling "rules of thumb" for everyday social norms. It reflects Western-centric perspectives and struggles to capture dynamic cultural variability.

**BBQ (Bias Benchmark for QA)** Parrish et al. (2022) Designed to reveal biases in question-answering, focusing on stereotypes against protected groups like gender and occupation-based biases.

**COCO-LM** Meng et al. (2021) Assesses contextual coherence by training models to correct and contrast corrupted text sequences.

**CrOWs Pair** Nangia et al. (2020a) Measures implicit stereotypical biases by comparing the likelihood assigned to stereotyped versus neutral sentences.

**Social Bias Frames** Sap et al. (2020) Analyzes social bias implications embedded in textual outputs, categorizing language along dimensions of bias and offense.

**X-FACTR** Jiang et al. (2020) Tests factual consistency in multilingual question-answering tasks, probing language models' world knowledge across languages.

**BOLD** Dhamala et al. (2021) Evaluates biases in open-ended generation around attributes like gender, race, and religion.

**Social IQA** Sap et al. (2019) Measures the understanding of everyday social interaction norms, requiring models to infer appropriate behavior in scenarios.

**BLEND** Myung et al. (2024) Assesses a model's ability to generate coherent narratives by blending multiple informational sources.

**M5** Schneider and Sitaram (2024) Benchmarks vision-language models on multilingual, multicultural tasks across diverse regions and modalities.

**SeaEval** Wang et al. (2024a) Evaluates LLMs' ability to reason about culturally sensitive scenarios across multiple languages.

**NorMAD** Rao et al. (2024) Measures culturally nuanced factual knowledge across different historical and political perspectives.

**CultureAtlas** Fung et al. (2024) Tests adaptability to cultural conversational norms across 100+ countries, although with significant representation imbalances.

**CIVICS** Pistilli et al. (2024) Examines LLMs' understanding of socially sensitive issues like LGBTQ+ rights, immigration, and surrogacy across five languages.

**MoralBench** Ji et al. (2024) Probes moral reasoning using culturally grounded dilemmas based on Moral Foundations Theory.

**Americas NLI** Ebrahimi et al. (2022) Expands natural language inference datasets to 10 Indigenous languages from the Americas, targeting low-resource evaluation.

**VLBiasBench** Wang et al. (2024b) Evaluates biases in multimodal vision-language models across a wide range of protected and intersectional categories.

### A.2 Culture Atlas

| Country | Sent. | CRR | Country | Sent. | CRR | Country | Sent. | CRR |
|---|---|---|---|---|---|---|---|---|
| United States | 42400 | 28.53 | India | 18400 | 12.38 | Italy | 8500 | 5.72 |
| China | 8100 | 5.45 | United Kingdom | 7200 | 4.85 | Indonesia | 6400 | 4.31 |
| Germany | 6200 | 4.17 | Japan | 5700 | 3.84 | Spain | 3500 | 2.36 |
| Thailand | 2900 | 1.95 | Canada | 2800 | 1.88 | Australia | 2800 | 1.88 |
| Israel | 2200 | 1.48 | Brazil | 2200 | 1.48 | Sweden | 1900 | 1.28 |
| Switzerland | 1900 | 1.28 | Tunisia | 1800 | 1.21 | Bulgaria | 1700 | 1.14 |
| Belgium | 1400 | 0.94 | Argentina | 1400 | 0.94 | Ukraine | 1300 | 0.87 |
| Ireland | 1200 | 0.81 | Greece | 1100 | 0.74 | South Africa | 1000 | 0.67 |
| Austria | 900 | 0.61 | Bangladesh | 800 | 0.54 | Viet Nam | 800 | 0.54 |
| Türkiye | 700 | 0.47 | Cambodia | 600 | 0.40 | Kenya | 600 | 0.40 |
| Armenia | 600 | 0.40 | United Arab Emirates | 600 | 0.40 | Lebanon | 500 | 0.34 |
| Algeria | 400 | 0.27 | Azerbaijan | 400 | 0.27 | Guinea-Bissau | 400 | 0.27 |
| Uruguay | 400 | 0.27 | Lithuania | 400 | 0.27 | Lesotho | 300 | 0.20 |
| Ghana | 300 | 0.20 | Chile | 300 | 0.20 | Zimbabwe | 300 | 0.20 |
| Hungary | 300 | 0.20 | South Sudan | 300 | 0.20 | Bhutan | 300 | 0.20 |
| Iraq | 300 | 0.20 | Zambia | 200 | 0.13 | Afghanistan | 200 | 0.13 |
| Uzbekistan | 200 | 0.13 | Georgia | 200 | 0.13 | Uganda | 200 | 0.13 |
| Trinidad and Tobago | 200 | 0.13 | Botswana | 200 | 0.13 | Bosnia and Herzegovina | 200 | 0.13 |
| Sudan | 200 | 0.13 | Belarus | 200 | 0.13 | Sri Lanka | 200 | 0.13 |
| Angola | 200 | 0.13 | Laos | 200 | 0.13 | Kyrgyzstan | 100 | 0.07 |
| Turkmenistan | 100 | 0.07 | Somalia | 100 | 0.07 | Latvia | 100 | 0.07 |
| Iran | 100 | 0.07 | Cameroon | 100 | 0.07 | Bahrain | 100 | 0.07 |
| Kiribati | 100 | 0.07 | Bolivia | 100 | 0.07 | Benin | 100 | 0.07 |
| Barbados | 100 | 0.07 | Honduras | 100 | 0.07 | Jordan | 100 | 0.07 |
| Kazakhstan | 100 | 0.07 | Burkina Faso | 100 | 0.07 | Bahamas | 0 | 0.00 |
| Antigua and Barbuda | 0 | 0.00 | Belize | 0 | 0.00 | Burundi | 0 | 0.00 |
| Suriname | 0 | 0.00 | Andorra | 0 | 0.00 | Syrian Arab Republic | 0 | 0.00 |
| Brunei Darussalam | 0 | 0.00 | Timor-Leste | 0 | 0.00 | Togo | 0 | 0.00 |
| Tonga | 0 | 0.00 | Kuwait | 0 | 0.00 | Cabo Verde | 0 | 0.00 |
| Tuvalu | 0 | 0.00 | Central African Republic | 0 | 0.00 | Chad | 0 | 0.00 |
| Grenada | 0 | 0.00 | Guatemala | 0 | 0.00 | Guinea | 0 | 0.00 |
| Guyana | 0 | 0.00 | Haiti | 0 | 0.00 | Yemen | 0 | 0.00 |
| Iceland | 0 | 0.00 | Albania | 0 | 0.00 | Jamaica | 0 | 0.00 |
| Liechtenstein | 0 | 0.00 | Liberia | 0 | 0.00 | Solomon Islands | 0 | 0.00 |

Table 1: Cultural Representation Rates (CRR) across 102 countries